Estimating the Causal Effect of Extracurricular Activities on Student Performance

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Abstract

This project investigates the causal relationship between extracurricular activity participation and student exam scores using observational data. We start with ordinay least squares (OLS) regression to estimate the average treatment effect (ATE) while controlling for observed covariates. We then explore potential treatment effect heterogeneity by including interaction terms between the treatment and selected covariates. Finally, we implement inverse probability weighting (IPW) based on propensity scores to further adjust for confounding bias. The results consistently indicate a positive association between extracurricular activities and exam scores, with estimated effects around 0.5 to 0.6 points higher for participants compared to non-participants. However, the magnitude of the estimated effect is sensitive to influential observations, although the overall conclusion of a positive effect remains robust. The exploration of treatment effect heterogeneity did not yield statistically significant interaction effects, suggesting that the benefit of extracurricular activities on exam scores does not vary strongly across different levels of motivation or study hours in this dataset. The propensity score diagnostics indicate that the IPW approach successfully balanced the covariates between treatment groups, lending credibility to the causal interpretation of the estimated effects under the unconfoundedness assumption.

1 Introduction

Understanding the factors that influence student performance is crucial for educators, policymakers, and students themselves. While factors such as family background, study habits, and school environment have been known to be associated with academic performance, distinguishing casuality from correlation remains a challenge.

Also, extra-curricular activity has been widely debated in terms of its impact on academic performance. Some argue that it enhances time management and social skills, while others suggest it may detract from study time. This project will explore the relationship between extra-curricular activities and student exam scores, controlling for various confounding factors.

The primary goal of this project is to estimate the **causal effect** of extra-curricular activities on student exam scores, while controlling for other factors that may influence both the likelihood of participating in extra-curricular activities and academic performance.

2 Data Description

The data comes from Kaggle's "Student Exam Performance Dataset Analysis" [1], which contains information on students's academic, lifestyle and socioeconomic factors.

Overall, the dataset includes 6,607 student observations and 20 columns, including 19 predictor variables and 1 response variable (Exam Score). Among all the predictors, we have 6 continuous variables (e.g., Hours Studied, Sleep Hours) and 13 categorical variables (e.g., Gender, School Type).

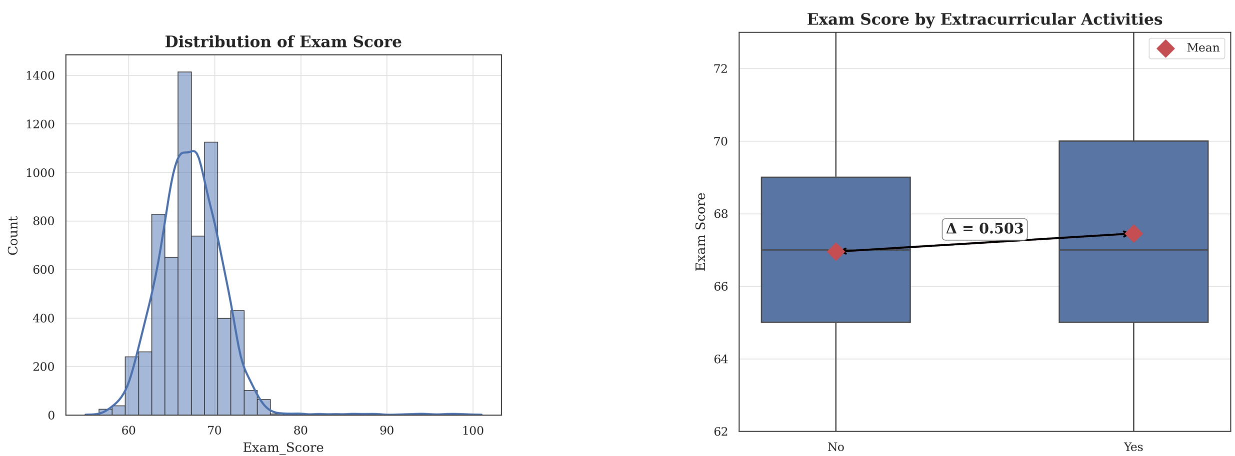
The brief summaries of the variables are given in Tables 5 and 6 in the Appendix. After removing the observations with missing values, we have 6,378 complete cases for analysis.

3 Exploratory Data Analysis

3.1 Key Variable Distribution

For this project, we are particularly interested in two key variables: **Extracurricular Activities** (the treatment variable) and **Exam Score** (the outcome variable). Figure 1a shows the distribution of exam scores, which appears to be centered around 67. It also shows a slight right skew, indicating that while most students score around the mean, there are some students who perform exceptionally well. Figure 1b illustrates the distribution of exam scores by extracurricular activity participation. For naîve comparison, we can see the mean difference in exam scores between students who participate in extracurricular activities and those who do not is around 0.503, suggesting that students who participate in extracurricular activities tend to have slightly higher exam scores on average.

Figure 1: Distribution of Key Variables



(a) Distribution of Exam Scores

(b) Distribution of Exam Scores by Extracurricular Activity Participation

3.2 Covariate Balance

To quantitatively assess the balance of covariates between the treatment and control groups, we compute the standardized mean differences (SMD) for all covariates. The formula for SMD is given by:

$$\text{SMD} = \frac{\bar{X}_T - \bar{X}_C}{\sqrt{\frac{S_T^2 + S_C^2}{2}}}$$

where $\bar{X}_T$ and $\bar{X}_C$ are the means of the covariate in the treatment and control groups, respectively, and S_T^2 and S_C^2 are the corresponding variances. For categorical variables, we first carry out one-hot encoding and then compute the SMD for each category. The SMD is a commonly used metric to evaluate the balance of covariates in observational studies, with values less than 0.1 generally indicating good balance.

Figure 5 shows the largest SMD is around 0.06, which suggests that the covariates are reasonably balanced between the treatment and control groups, although there may still be some residual confounding that needs to be addressed in the further analysis. We choose to keep all the covariates in the model to control for potential confounding bias, as they are all theoretically relevant to the relationship between extracurricular activities and exam scores.

4 Methodology

4.1 Causal Inference Assumption

Because the data are observational rather than experimental, identification of the causal effect of extracurricular activities on exam scores relies on additional assumptions, as discussed in [2].

- **Unconfoundedness**: We assume that, conditional on the observed covariates, the assignment of treatment (participation in extracurricular activities) is independent of the potential outcomes (exam scores). This means that there are no unobserved confounders that affect both the treatment and the outcome.

  $$Y(0), Y(1) \perp T \mid X$$

  where $Y(0)$ and $Y(1)$ are the potential outcomes under control and treatment, respectively, T is the treatment indicator, and X represents the observed covariates.

- **Overlap (Positivity)**: We assume that there is a positive probability of receiving each treatment level for all values of the covariates. In other words, for any combination of covariates, there are students who participate in extracurricular activities and students who do not.

  $$0 < P(T = 1 \mid X) < 1$$

- **Stable Unit Treatment Value Assumption (SUTVA)**: We assume that the potential outcomes for any student are unaffected by the treatment assignment of other students. This means that there are no interference between units and no hidden variations of treatment.

4.2 Estimation Methods

Based on the above assumptions, we plan to use the following methods to estimate the causal effect of extracurricular activities on exam scores. We will compare the results from these methods to assess the robustness of our findings.

- **Outcome Regression (OLS)**: We will fit an ordinary least squares regression model controlling for all observed covariates to estimate the average treatment effect (ATE).
- **Treatment Effect Heterogeneity**: We will explore potential heterogeneity in treatment effects by including interaction terms between the treatment and selected covariates.
- **Propensity Score and Weighted Least Squares (WLS)**: We will estimate the propensity score using logistic regression and then apply inverse probability weighting to create a pseudo-population where the distribution of covariates is balanced between the treatment and control groups. We will then use weighted least squares regression to estimate the ATE.

4.3 Diagnosis

The model diagnostics consist of two parts, one for linear models and one for propensity score models.

- For OLS, we will check the residuals to assess the assumptions of linearity and homoscedasticity. For influential observations, we will compute leverage and Cook's distance to identify potential outliers and influential points. If we find influential observations, we will refit the model after removing them and compare the results to assess the robustness of our findings. We will also check for multicollinearity among the covariates using variance inflation factors (VIF).
- For propensity score models, we will evaluate the balance of covariates after weighting by comparing the standardized mean differences before and after weighting. We will also check the distribution of propensity scores to ensure there is sufficient overlap between the treatment and control groups.

5 Outcome Regression (OLS)

5.1 Model Specification

Under the assumption that all relevant confounding variables are observed and the linear model is correctly specified, the average treatment effect (ATE) can be estimated using ordinary least squares (OLS). The model is specified as:

$$Y_i = \alpha_0 + \alpha_1 T_i + X_i^T \alpha + \epsilon_i$$

where α_1 represents the main effect of extracurricular activities on exam scores, conditional on covariates X_i.

5.2 Results

5.2.1 Main Effect

Table 1 reports the OLS estimate of the treatment effect. The estimated coefficient for extracurricular activities is 0.562 (95% CI: [0.458, 0.666], $p < 0.001$), indicating that, holding other observed covariates constant, students who participate in extracurricular activities score on average 0.56 points higher than those who do not. The model explains a substantial portion of the variation in exam scores ($R^2 = 0.722$).

Table 1: OLS Estimate of the Main Effect of Extracurricular Activities on Exam Scores

Effect	Std. Err.	p-value	CI Lower	CI Upper	R^2	Adj. R^2	N
0.562	0.053	<0.001	0.458	0.666	0.722	0.720	6378

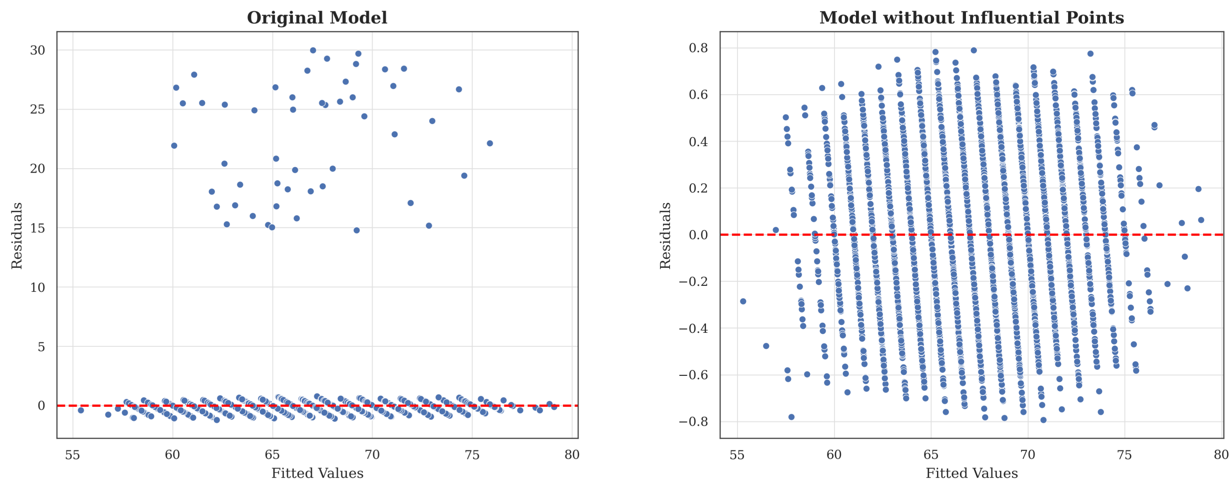
5.2.2 Model Diagnostics

To assess the validity of the OLS assumptions, we examine residual patterns and potential influence of individual observations.

Residual Analysis The residual-versus-fitted plot in Figure 2 (left panel) reveals a distinct group of observations with large positive residuals, suggesting that the model may not fit well for a subset of the data. This pattern may arise from either outliers or model misspecification.

Influence Diagnostics To further investigate, we compute influence measures based on leverage, and Cook's distance. Using common thresholds ($4/N$ for Cook's distance and $\frac{2p}{N}$ for leverage), approximately 0.9% of observations are identified as potentially influential. As shown in Figure 3, only a small number of observations exhibit relatively large Cook's distance values, with the maximum around 0.05, indicating moderate influence.

Robustness to Influential Observations To evaluate the impact of those influential observations, we refit the model after excluding the identified influential points. The resulting residual plot (Figure 2, right panel) shows a more symmetric and random distribution around zero, suggesting improved model fit and more plausible homoskedasticity.



(a) Original OLS model

(b) After removing influential points

Figure 2: Residual diagnostics before and after removing influential observations in OLS.

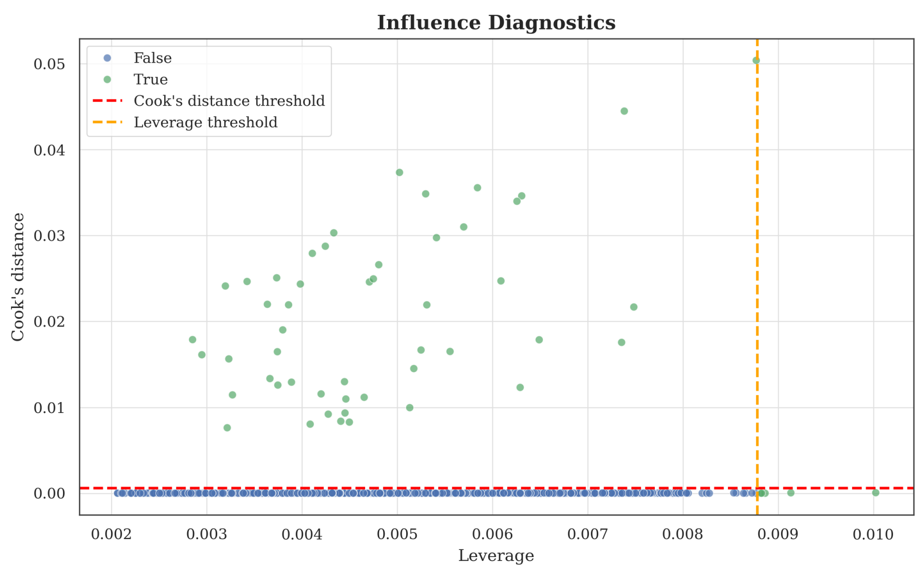


Figure 3: Influence diagnostics for OLS model based on leverage and Cook's distance. The dashed lines indicate common thresholds for identifying influential observations.

Table 2 reports the updated coefficient estimates. The estimated main effect remains positive and statistically significant, with a slightly reduced magnitude (0.506 compared to 0.562) and a narrower confidence interval. This indicates that while influential observations affect the magnitude of the estimate, the overall conclusion is robust. For the downstream exploration, we keep the whole data for analysis since first we have little prior information about the influential observations and the ratio is quite small and acceptable. However, we will also report the results after removing influential observations for comparison and robustness check.

Multicollinearity We assess multicollinearity using variance inflation factors (VIF). All VIF values are below 5, with a maximum of 2.826, indicating no evidence of severe multicollinearity. This suggests that the coefficient estimates are stable and not overly sensitive to small perturbations in the data.

Table 2: OLS Estimate After Removing Influential Observations

Effect	Std. Err.	p-value	CI Lower	CI Upper	R^2	Adj. R^2	N
0.506	0.008	<0.001	0.490	0.521	0.991	0.991	6323

6 Treatment Effect Heterogeneity

6.1 Model Specification

To explore treatment effect heterogeneity, we extend the model by including interaction terms between treatment and selected covariates. This allows the treatment effect to vary across subgroups while maintaining the same identification assumptions.

$$Y_i = \alpha_0 + \alpha_1 T_i + X_i^T \alpha + (T_i \cdot X_i)^T \gamma + \epsilon_i$$

where γ captures the heterogeneity in treatment effects across different levels of covariates. After exploration, we select two covariates (**Hours studied** and **Motivation level**) to include in the interaction terms, as they show the largest imbalance between treatment and control groups qualitatively and are also theoretically relevant to the relationship between extracurricular activities and exam scores.Visualizations of the interaction patterns are provided in Figures 6 and 7 in the Appendix.

6.2 Result

Following the same way as the main effect model, we provide the coefficient estimates for the interaction model both with and without influential observations, followed by the model diagnostics for residual patterns and multicollinearity.

6.3 Interpretation

Details of the coefficient estimates are shown in Table 3. For the categorical variable motivation level, we use the high motivation group as the reference category. For the continuous variable hours studied, the coefficient represents the change in treatment effect for each additional hour studied.

- **Baseline treatment effect.** In the interaction model, the coefficient on extracurricular activities represents the treatment effect for the reference group, i.e., students with high motivation (the omitted category) and zero study hours. The estimated effect is 0.921 (95% CI: [0.505, 1.337], $p < 0.001$).

  However, the interpretation at zero study hours is not substantively meaningful, as the minimum value in the dataset is 1. Therefore, this coefficient should be viewed primarily as a baseline for interpreting interaction effects rather than a standalone causal estimate.

  After removing influential observations, the baseline effect decreases to 0.489 (95% CI: [0.425, 0.553], $p < 0.001$), indicating that the magnitude of the estimated effect is sensitive to influential data points, although the positive effect remains robust.

- **Interaction with motivation level.** The interaction terms capture how the treatment effect differs across motivation levels relative to the high-motivation group. The estimated interaction coefficients are -0.250 for low motivation and -0.190 for medium motivation. This implies that the treatment effect is smaller for students with lower motivation.

  For example, holding study hours fixed at zero (for comparison), the implied treatment effects are:

  - High motivation: 0.921
  - Medium motivation: $0.921 - 0.190 = 0.731$
  - Low motivation: $0.921 - 0.250 = 0.671$

While these point estimates suggest that highly motivated students may benefit more from extracurricular activities, both interaction terms are statistically insignificant at the 5% level. Therefore, there is no strong empirical evidence that the treatment effect varies systematically by motivation level.

After removing influential observations, the interaction effects shrink substantially toward zero and remain statistically insignificant, further indicating that the evidence for heterogeneity by motivation level is weak and not robust.

- **Interaction with study hours.** The interaction between extracurricular participation and study hours is estimated as -0.009 ($p = 0.287$). This implies that the treatment effect varies with study hours according to:

$$\text{Heterogeneous Treatment Effect} = 0.921 - 0.009 \times \text{Hours Studied}.$$

This suggests a negative relationship between study time and the marginal benefit of extracurricular activities, potentially reflecting a substitution effect between academic effort and extracurricular engagement.

However, the interaction term is not statistically significant, indicating that there is no strong evidence that the treatment effect varies with study hours. After removing influential observations, the interaction coefficient becomes even closer to zero and remains insignificant, reinforcing the conclusion that the heterogeneous effect is not supported by the data.

Table 3: Heterogeneous Treatment Effects with and without Influential Observations

	With influentials			Without influentials		
Term	Coef.	95% CI	p-value	Coef.	95% CI	p-value
Extracurricular Activities	0.921	[0.505, 1.337]	<0.001	0.489	[0.425, 0.553]	<0.001
× Motivation (Low)	-0.250	[-0.551, 0.052]	0.104	-0.040	[-0.086, 0.007]	0.094
× Motivation (Medium)	-0.190	[-0.465, 0.085]	0.175	-0.004	[-0.046, 0.038]	0.858
× Hours Studied	-0.009	[-0.027, 0.008]	0.287	0.002	[-0.001, 0.004]	0.270

Model Diagnostics We carry out the same diagnostic checks for the interaction model as we did for the main effect model. After moving the influential observations, the residual patterns show improved symmetry and reduced heteroskedasticity. We see high VIF in the interaction terms with treament variable. But for other covaraites, the VIF values are still below 5, indicating no severe multicollinearity.

7 Propensity Score and WLS

7.1 Model Specification

Propensity score measures the probability of receiving treatment given the observed covariates. By estimating the propensity score, we can create a weighted dataset that mimics a randomized experiment, where the distribution of covariates is balanced between the treatment and control groups. In practice, we use logistic regression to estimate the propensity score and then apply the weights to estimate the ATE using weighted least squares (WLS) regression.

The propensity score model is specified as:

$$P(T_i = 1 \mid X_i) = \frac{\exp(\beta_0 + X_i^T \beta)}{1 + \exp(\beta_0 + X_i^T \beta)}$$

The WLS model is then specified as:

$$Y_i = \alpha_0 + \alpha_1 T_i + X_i^\top \alpha + \epsilon_i$$

where the weights are defined as:

$$w_i = \frac{T_i}{P(T_i = 1 \mid X_i)} + \frac{1 - T_i}{1 - P(T_i = 1 \mid X_i)}$$

7.2 Result

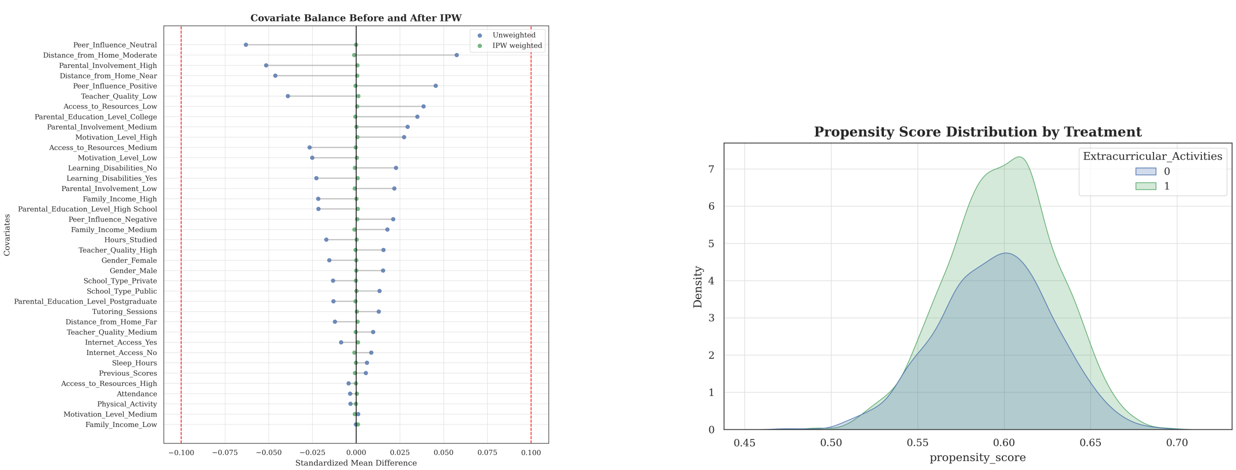
7.2.1 Propensity Score Evaluation

- **Model Fit**: With all the covariates included, the logistic regression model for the propensity score shows poor fit, with a pseudo-R^2 of 0.003 and all the covariates being statistically insignificant at the 5% level. This suggests that the observed covariates do not strongly predict treatment assignment, which may indicate that the unconfoundedness assumption is more plausible in this context.

- **Covariate Balance**: After applying the inverse probability weights, we reassess the balance of covariates between the treatment and control groups. Figure 4a shows all the standardized mean differences becomes much smaller after weighting compared with the unweighted sample. Althought for unweighted sample, the largest SMD is already below the common threshold of 0.1, the weighted sample shows even better balance with all SMDs close to zero, indicating that the IPW approach has successfully balanced the covariates between the treatment and control groups.

- **Overlap Assessment**: We examine the distribution of propensity scores to ensure sufficient overlap between the treatment and control groups. The distribution in Figure 4b shows the propensities are well distributed across the treatment and control groups.

Figure 4: Propensity Score Diagnostics



(a) Covariate Balance Before and After IPW

(b) Distribution of Propensity Scores by Treatment Group

The poor fit of the propensity score model gives us some confidence that the unconfoundedness assumption may hold, as the treatment assignment does not appear to be strongly driven by the observed covariates. The improved balance after weighting also supports the validity of the IPW approach for estimating the causal effect.

7.2.2 ATE Estimation

We apply the inverse probability weights to estimate the ATE using weighted least squares regression. The results are shown in Table 4. The estimated treatment effect is 0.562 (95% CI: [0.462, 0.662], $p < 0.001$) when including all observations, which is consistent with the OLS estimate. After removing influential observations, the estimated effect decreases to 0.505 (95% CI: [0.490, 0.521], $p < 0.001$), indicating that influential points have a noticeable impact on the magnitude of the estimated effect, although the positive association remains robust.

The 95% confidence intervals in baseline outcome regression and IPW are quite similar, even the same after removing influential observations, which suggests that the IPW approach is providing a similar estimate of the treatment effect as the OLS model, lending credibility to the causal interpretation under the unconfoundedness assumption.

Table 4: Inverse Probability Weighted (IPW) Estimates with and without Influential Observations

Model	Effect	Std. Err.	95% CI	p-value	N
With influential observations	0.562	0.051	[0.462, 0.662]	<0.001	6378
Without influential observations	0.505	0.008	[0.490, 0.521]	<0.001	6323

8 Key Findings and Interpretation

The results from both the OLS and IPW analyses consistently indicate a positive association between extracurricular activity participation and exam scores, with estimated effects around 0.5 to 0.6 points higher for students who participate in extracurricular activities compared to those who do not, after controlling for observed covariates. The robustness checks, including the removal of influential observations, show that while the magnitude of the estimated effect is sensitive to certain data points, the overall conclusion of a positive effect remains consistent. The exploration of treatment effect heterogeneity did not yield statistically significant interaction effects, suggesting that the benefit of extracurricular activities on exam scores does not vary strongly across different levels of motivation or study hours in this dataset. The propensity score diagnostics indicate that the IPW approach successfully balanced the covariates between treatment groups, lending credibility to the causal interpretation of the estimated effects under the unconfoundedness assumption.

Based on these findings, we can cautiously conclude that participation in extracurricular activities is associated with improved academic performance as measured by exam scores. However, it is important to acknowledge the limitations of the analysis, including potential unobserved confounding and the observational nature of the data, which may affect the generalizability of the results.

References

[1] Kaggle Contributors. Student exam performance dataset analysis. Kaggle, 2026.

[2] Peng Ding. A First Course in Causal Inference, October 2023. arXiv:2305.18793 [stat].

Appendix

A Data Summary Tables

Here we present the summary statistics for both continuous and categorical variables in the original dataset.

Table 5: Summary Statistics for Continuous Variables

Variable	Mean	Std. Dev.	Min	Max	Null Count	Null Rate
Hours_Studied	19.98	5.99	1.00	44.00	0	0.000
Attendance	79.98	11.55	60.00	100.00	0	0.000
Sleep_Hours	7.03	1.47	4.00	10.00	0	0.000
Previous_Scores	75.07	14.40	50.00	100.00	0	0.000
Tutoring_Sessions	1.49	1.23	0.00	8.00	0	0.000
Physical_Activity	2.97	1.03	0.00	6.00	0	0.000
Exam_Score	67.24	3.89	55.00	101.00	0	0.000

Table 6: Summary Statistics for Categorical Variables

Variable	No. of Categories	Mode	Null Count	Null Rate
Parental_Involvement	3	Medium	0	0.000
Access_to_Resources	3	Medium	0	0.000
Extracurricular_Activities	2	Yes	0	0.000
Motivation_Level	3	Medium	0	0.000
Internet_Access	2	Yes	0	0.000
Family_Income	3	Low	0	0.000
Teacher_Quality	4	Medium	78	0.012
School_Type	2	Public	0	0.000
Peer_Influence	3	Positive	0	0.000
Learning_Disabilities	2	No	0	0.000
Parental_Education_Level	4	High School	90	0.014
Distance_from_Home	4	Near	67	0.010
Gender	2	Male	0	0.000

B Standardized Mean Differences for All Covariates

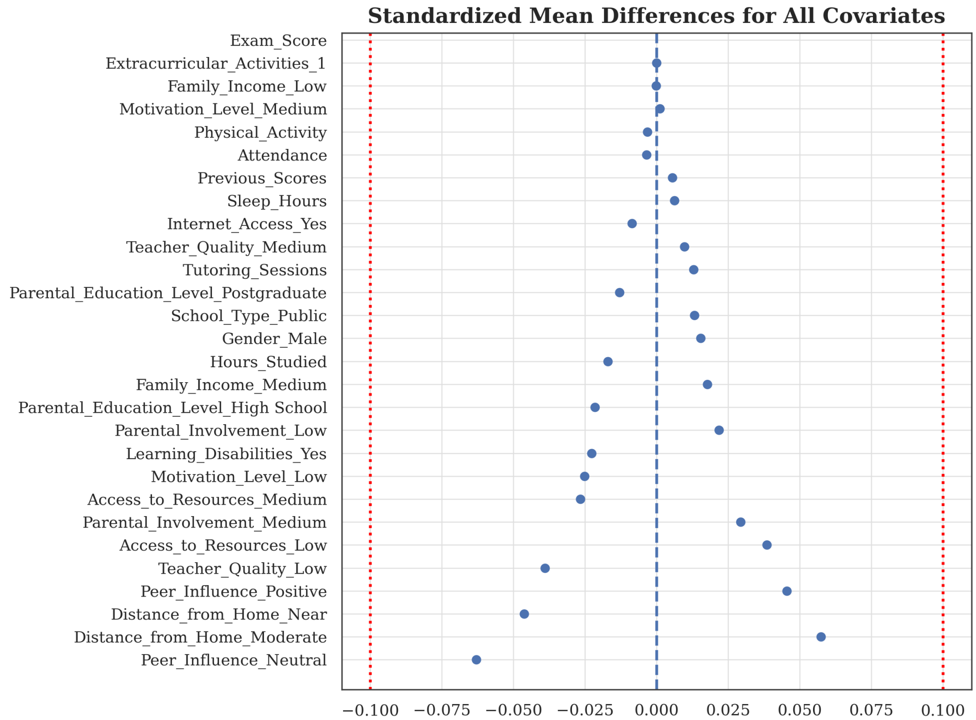


Figure 5: Standardized Mean Differences for All Covariates

C Interaction Plots

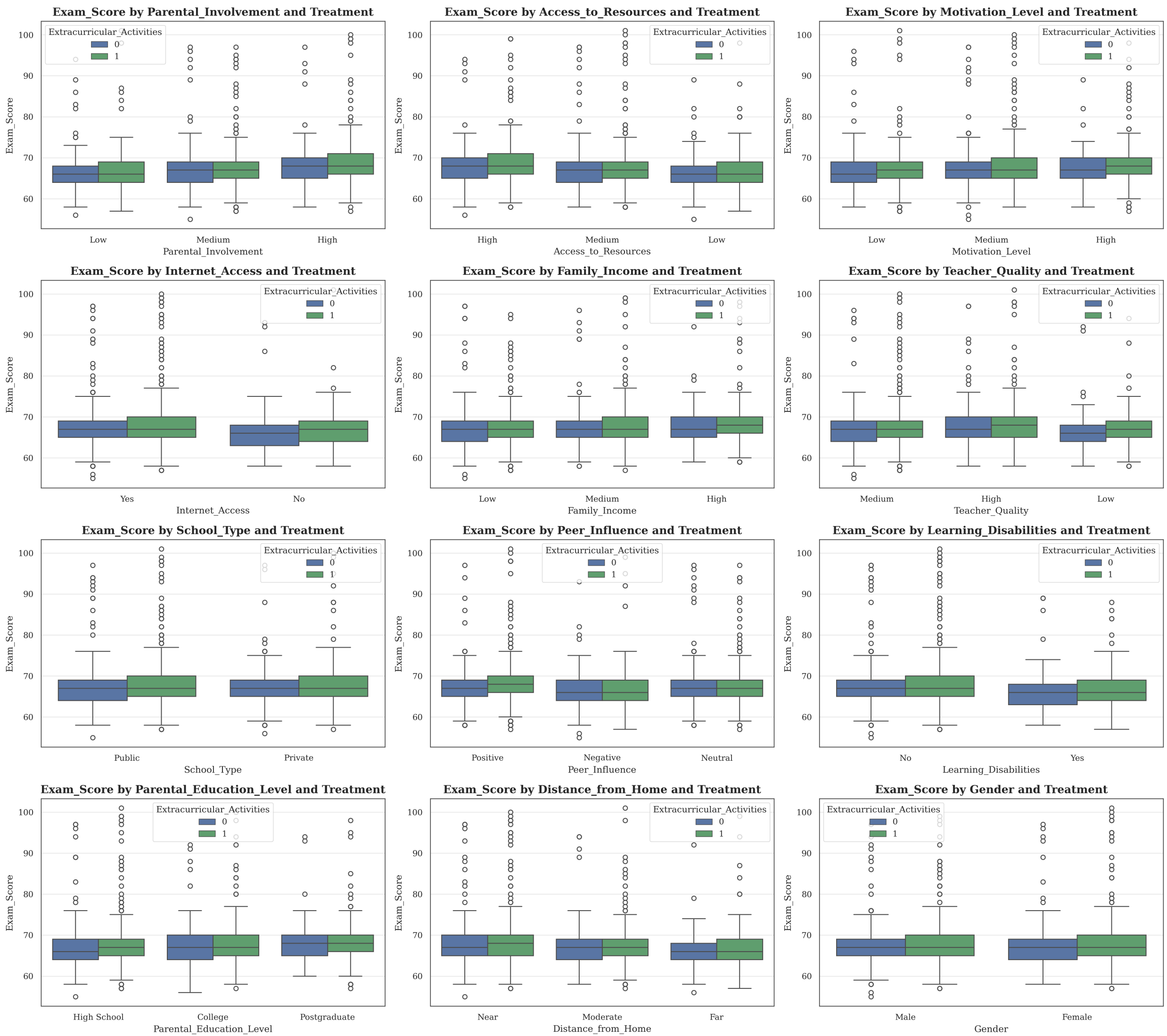


Figure 6: Interaction effects between treatment and categorical covariates.

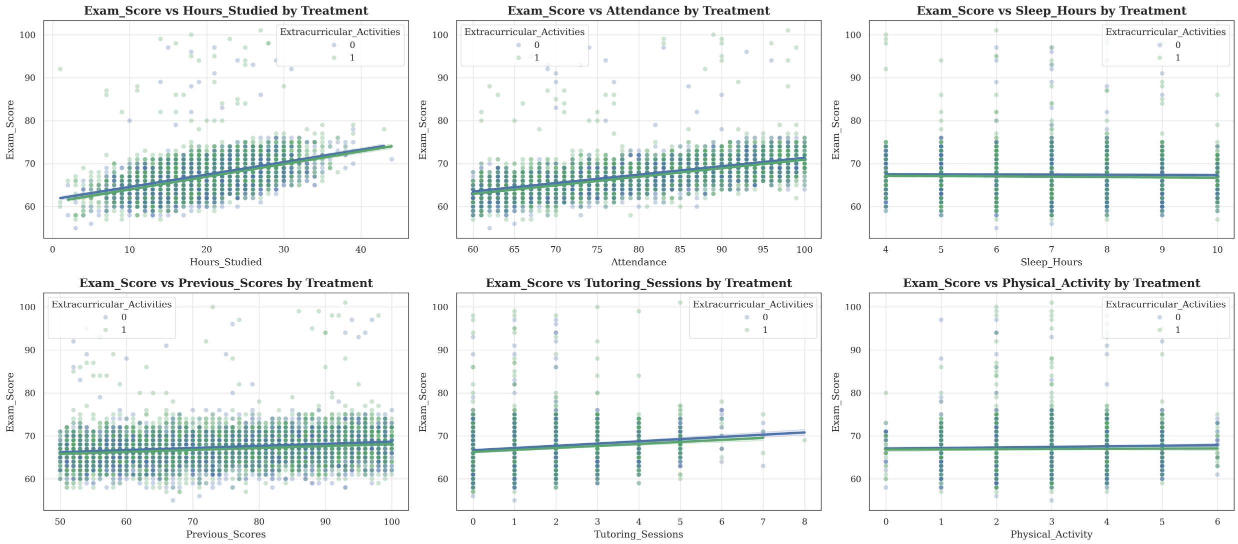


Figure 7: Interaction effects between treatment and continuous covariates.